DATA ITEM DESCRIPTION
Title: MATERIALS AND PROCESSES SUBSTANTIATING DATA AND ANALYSIS TEST PLAN

Number: DI-SESS-82236A
AMSC Number: N10401
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20230510
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2023-020

Use/relationship: The Materials and Processes Substantiating Data and Analysis Test Plan identifies the approach for documenting test plans and procedures that will substantiate the materials and processes used to develop materiel. The plan will be used to assess the materiel development approach to ensure system performance requirements are met.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described by the contract SOW.

This DID interfaces with DI-SESS-82237A, Materials and Processes Substantiating Data and Analysis Test Report.

This DID supersedes DI-SESS-82236.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. Contractor format acceptable.

3. Content. The Materials and Processes Substantiating Data and Analysis Test Plan shall contain the following:

3.1. Materials and Processes Evaluation: Describe the plan used to evaluate all proposed materials and processes to be used on the system for their ability to meet system performance requirements.

3.2. Testing Methods for Material Properties:

3.2.1. Describe plans for all research and development, testing, and fabrication studies to be accomplished with respect to materials and processes development.

3.2.2. Describe plans to conduct materials testing.

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3.2.3. Describe the tasks to be performed for the evaluation and substantiation of material properties.

3.2.4. Describe plans for developing non-destructive test accept/reject criteria, materials processing effects and parameters, material fabrication techniques, and manufacturing procedures.

3.2.5. Describe program specific test objectives, test methods, test procedures, test conditions, and test matrices.

3.2.6. Describe plans to validate the methods to be used in the testing and evaluation of materials.

3.2.7. Describe the plan for materials testing and evaluation to ensure that materials and processes are characterized in a statistically significant manner relative to the design application.

3.3. Material Properties Testing Plans and Procedures:

3.3.1. Describe the plans to use a testing and evaluation strategy supports the building block approach (coupon, element, sub component). Describe the rationale for using approaches other than the building block approach.

3.3.2. Describe the plan for the implementation of procedures used for monitoring and controlling processes critical to satisfying part design requirements.

3.3.3. Describe the plan for identifying, monitoring, controlling, and analyzing key process parameters.

3.3.4. Describe the process for determining and evaluating process capability, including management and reduction of process variation.

3.3.5. Describe the methodology to be used to determine and substantiate thermal exposure time histories.

3.3.6. Describe the plan to substantiate composite sandwich construction properties, including those analytically derived from sandwich constituent properties to permit for the statistical determination of the material properties.

3.3.7. Describe the methodology that will be used for process verification testing of test coupons (representative of the production part) and their associated acceptance criteria.

3.4. Material Properties Test Schedule: Describe a materials and processes testing schedule that includes the characterization, testing, and analysis planned with respect to critical path milestones against the contractor’s integrated master schedule.

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3.5. Material Properties Substantiation:

3.5.1. Describe the methods that will be used to substantiate material properties by using standardized documentation, analyses, tests, and inspections to characterize the following properties: static strength, fatigue strength, modulus, effect of stress concentrations, fracture toughness, crack growth, impact resistance, effects of defects, resistance to environmentally-assisted cracking, wear, general corrosion, pitting, stress corrosion cracking, exfoliation, galvanic corrosion, crevice corrosion, filiform corrosion, moisture absorption, chemical, solvent, fuel, and lubricant exposure; hydrolytic instability, thermal exposure, electromagnetic compatibility, adhesion strength, peel resistance, dielectric constant, resistivity, glass transition temperature, and erosion.

3.5.2. Describe the approach for substantiating the effects of environmental exposure and operating environment on material properties.

3.5.3. Describe the methods that will be used to substantiate materials specifications selected for protective finishes, protective coatings, and sealants for corrosion protection.

3.6. Material Allowables Development:

3.6.1. Describe the design allowable philosophy that will be used to develop the system materials, and document how the data will be generated and used to determine material properties.

3.6.2. Describe the analytical procedures that will be used to calculate static and dynamic material allowables.

3.6.3. Describe the static material allowables methodology in statistical terms of confidence levels and exceedance values.

3.6.4. Describe the methodology used to substantiate the impact of inorganic finish systems on material allowables.

3.6.5. Describe the approach to generate composite laminate data (including data analytically derived from single-ply properties) to permit for a statistical determination of the laminate properties.

3.6.6. Describe a plan to substantiate the system materials' fracture toughness values used in the residual strength analysis of metallic parts will be the minimum substantiated fracture toughness values.

End of DI-SESS-82236A

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